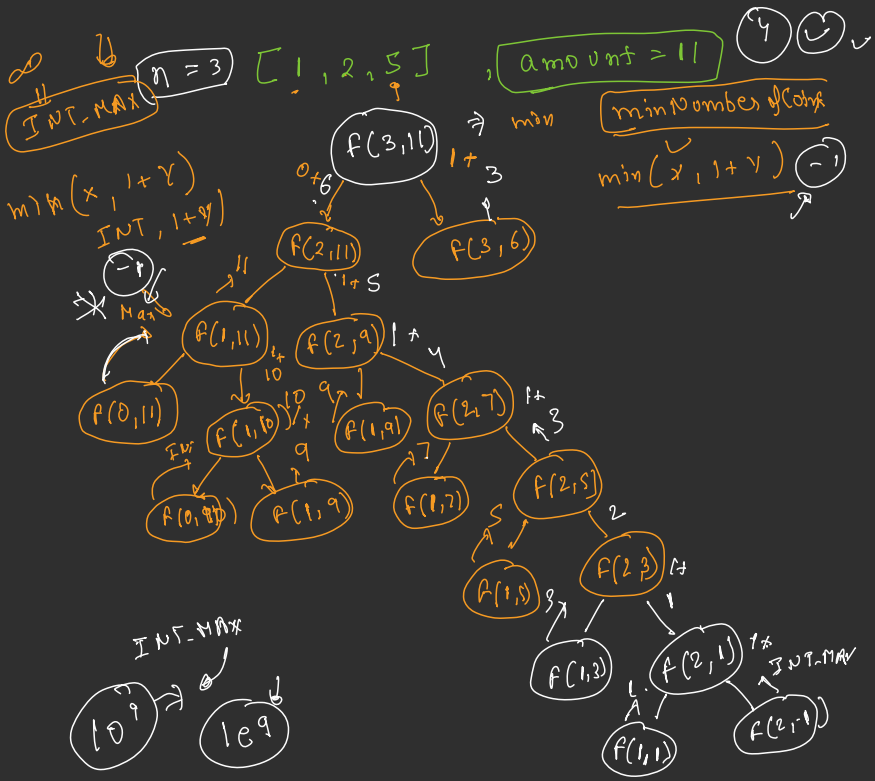


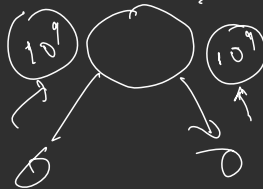




$1 + 10^9$

[3, 6, 15]

amount = 2



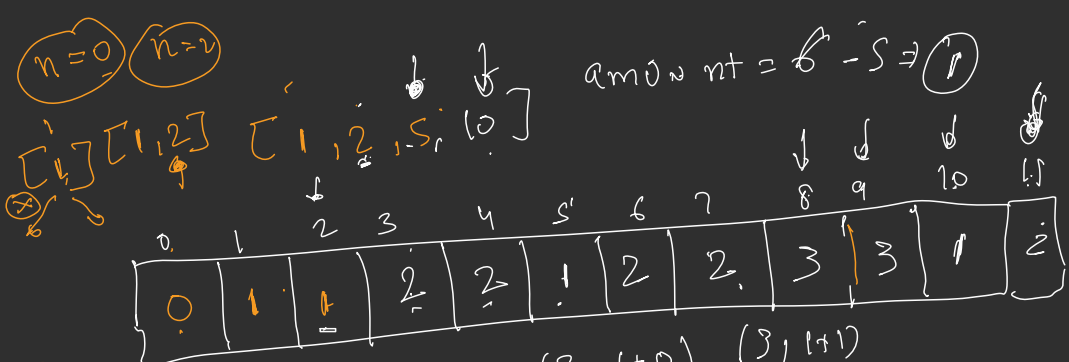
```
int findMinCoin(int n, int amount, arr)
{
    if (amount == 0)
        return 0;
    if (n == 0 || amount < 0)
        return 10^9;
    min ( findMinCoin(n-1, amount, arr)
          1 + findMinCoin(n, amount - arr[n-1], arr) )
}
```

$n \times \text{amount}$
 $n \geq 3, \quad [1, 2, 5]$
 $dp[i][j] = \min(dp[i-1][j], 1 + dp[i][j - \text{coins}[i-1]])$

$\uparrow$
 n, i
 $\downarrow$

0	109	109	109	109	109	109	109	709	109	109	109
10											
20											
30											

$\leftarrow$ amount $\rightarrow$
 j



$i-1$ $i+0$

$(3, 4+0), (3, 4+1)$

$[1, 2] \Rightarrow S$

$\Sigma 21 \approx 3$

If only one is available,

How much min coin is required to achieve amount

$n = 65$ $[1, 4, 9, 16, 25, 36]$

$i=1, iL=3, i++$

$n=12$

$[1, 4, 9]$

array

recursion

coin problem

coin $\rightarrow$ [1, 4, 9, 16, 25, ...] $\rightarrow$ array

$n=0$

0 1

$n=12$

answer
9 10 11 12

0	1	2	3	4	5	6	7	8	9	10	11	12
0	1	2	3	1	2	3	4	2	1	2	3	3

minPerfectSquare(9) ? $\rightarrow$ [1, 4, 9]

$\rightarrow$ minSqr(0) $\rightarrow$ 4, 3, 4

2 $\rightarrow$ minPerfectSquare(8) 2 $\rightarrow$ minSqr(5)

target = 4

